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Influence of Respiratory Frequency of Slow-Paced Breathing on Vagally-Mediated Heart Rate Variability

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Abstract

Breathing techniques, particularly slow-paced breathing (SPB), have gained popularity among athletes due to their potential to enhance performance by increasing cardiac vagal activity (CVA), which in turn can help manage stress and regulate emotions. However, it is still unclear whether the frequency of SPB affects its effectiveness in increasing CVA. Therefore, this study aimed to investigate the effects of a brief SPB intervention (i.e., 5 min) on CVA using heart rate variability (HRV) measurement as an index. A total of 75 athletes (22 female; $M_{age} = 22.32$; age range = 19–31) participated in the study, attending one lab session where they performed six breathing exercises, including SPB at different frequencies (5 cycles per minute (cpm), 5.5 cpm, 6 cpm, 6.5 cpm, 7 cpm), and a control condition of spontaneous breathing. The study found that CVA was significantly higher in all SPB conditions compared to the control condition, as indexed by both root mean square of the successive differences (RMSSD) and low-frequency HRV (LF-HRVms^2). Interestingly, LF-HRVms^2 was more sensitive in differentiating the respiratory frequencies than RMSSD. These results suggest that SPB at a range of 5 cpm to 7 cpm can be an effective method to increase CVA and potentially improve stress management and emotion regulation in athletes. This short SPB exercise can be a simple yet useful tool for athletes to use during competitive scenarios and short breaks in competitions. Overall, these findings highlight the potential benefits of incorporating SPB into athletes' training and competition routines.

Keywords Cardiac coherence · Deep breathing · Heart rate variability · Respiration · RMSSD

Introduction

In a recent meta-analysis, the use of breathing techniques as performance-enhancing interventions was found to be increasing among athletes (Laborde et al., 2022b). One of these breathing techniques is slow-paced breathing (SPB), which is suggested to positively influence a range of outcomes in athletes, such as stress management, emotion regulation, endurance, well-being, and sleep quality (Borges et al., 2021; Jimenez Morgan & Molina Mora, 2017; Laborde et al., 2022b; Pagaduan et al., 2020, 2021). One approach for athletes using SPB is to identify the ideal respiratory frequency for each individual, the so-called resonance frequency (Lehrer & Gevirtz, 2014; Shaffer & Meehan, 2020; Vaschillo et al., 2002). While having some merits, this approach remains time-consuming, and requires the use of specific apparatus. Additionally, there are no clear conclusions regarding outcomes when using SPB at the individual resonance frequency or a standard frequency, such as 6 cycles per minute (cpm) (Lehrer et al., 2020). Consequently,

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adopting a standard frequency for training with athletes may therefore represent a first step in SPB training. To date there is a lack of research regarding the effects of different SPB respiratory frequencies and the effect on cardiac vagal activity (CVA), therefore the current study aimed to address this gap.

SPB is a breathing technique that involves slowing down respiratory frequency, usually below 10 cpm (Laborde et al., 2022a; Russo et al., 2017; Zaccaro et al., 2018), while spontaneous breathing frequency is typically between 12 and 20 cpm (Sherwood, 2006; Tortora & Derrickson, 2014). The term “paced” reflects the controlled duration of the inhalation and exhalation phases. There are many mechanisms that are suggested to be activated by SPB (Gerritsen & Band, 2018; Noble & Hochman, 2019; Sevoz-Couche & Laborde, 2022). These include the improvement of baroreflex sensitivity, the increase of amplitude of respiratory sinus arrythmia (RSA - reflecting the increase of heart rate with inhalation and its decrease with exhalation (Eckberg, 1983), the stimulation of pulmonary stretch receptors, and the stimulation of brain networks involved in emotion regulation (Lehrer & Gevirtz, 2014; Mather & Thayer, 2018; Noble & Hochman, 2019; Sevoz-Couche & Laborde, 2022). All of these mechanisms are directed towards the involvement of the vagus nerve (Gerritsen & Band, 2018; Kromenacker et al., 2018; Sevoz-Couche & Laborde, 2022), the main nerve of the parasympathetic nervous system (Brodal, 2016). A non-invasive method of indexing the activity of the vagus nerve is via heart rate variability (HRV). HRV represents the change in the time intervals between successive heartbeats (Berntson et al., 1997; Laborde et al., 2017b; Malik, 1996) and cardiac functioning regulated via vagus nerve is referred to as cardiac vagal activity (CVA) (Berntson et al., 1997; Laborde et al., 2017a; Malik, 1996). Parameters reflecting CVA are described under the umbrella term vagally-mediated HRV (vmHRV).

vmHRV can be increased via SPB, as shown in a recent meta-analysis (Laborde et al., 2022a), therefore vmHRV is of great interest in the current study. vmHRV measurements that are often used include the root mean square of successive differences (RMSSD) in the time-domain, and the absolute power of high-frequency HRV (HF-HRVms^2) in the frequency-domain. For HFms^2 however, this interpretation only applies when breathing rate is between 9 and 24 cpm (Berntson et al., 1997), while the interpretation of RMSSD is not strongly influenced by respiratory frequency (Penttila et al., 2001). It is important to note that during SPB, there is a shift of CVA from HF-HRVms^2 to low-frequency (LF)-HRVms^2, as shown by a pharmacological blockade study (Kromenacker et al., 2018). Therefore, in the current study we will focus on two vmHRV outcome variables to index CVA during SPB, RMSSD and LF-HRVms^2.

SPB has been used with athletes, mainly with the help of a biofeedback system, enabling the individual to see the direct impact of SPB on psychophysiological variables, such as heart rate, HRV, or the respiratory frequency (Jimenez Morgan & Molina Mora, 2017; Laborde et al., 2021b; Laborde et al., 2022a; Pagaduan et al., 2020, 2021). However, it has been shown that the majority of SPB benefits can also be achieved without a biofeedback system (Laborde et al., 2021b; Wells et al., 2012). This makes SPB far more accessible to athletes as a performance tool given that it does not require any additional material besides a breathing pacer, therefore this is the method adopted in the current study.

There are a range of outcomes improved by a single session SPB without biofeedback in athletes (Borges et al., 2021; Laborde et al., 2022b). At the physiological level, a single SPB session can increase athletes’ CVA (You et al., 2021a). On the other hand, a period of adjustment to slow breathing may be required for subjective variables, such as perceived stress and emotional valence, as individuals new to the technique may experience some discomfort when slowing their breathing frequency (You et al., 2021c). A single session of SPB was found to help athletes improve executive functions (Laborde et al., 2021a), including after physical exertion (Laborde et al., 2019b). The latter results are in line with the positive effects of SPB in fostering recovery in athletes during repeated sprint ability training (Pelka et al., 2017) and after submaximal aerobic exercise (Perez-Gaido et al., 2021). Furthermore, a 2022 study by Conlon and colleagues examined the effects of SPB on pressurized pistol shooting performance. Findings in this study were mixed, showing some improvements at the level of psychological response (decreased cognitive anxiety and perceived stress), but not regarding performance as measured by shooting accuracy (Conlon et al., 2022). Furthermore, longer term SPB practice appears to have longer lasting benefit. Specifically, practicing SPB for 15 min over 30 days led to improvements in CVA during the night and subjective sleep quality (Laborde et al., 2019a).

Several aspects of SPB can be manipulated in research, such as its duration, the inhalation/exhalation ratio, and respiratory frequency (for a more extensive overview about potential moderators of the effects of SPB, see the recent meta-analysis from Laborde et al., 2022a). An investigation into the dose-response relationship showed no differences on CVA during and immediately after a single SPB session where the duration of SPB was manipulated (5 min, 10 min, 15 min, 20 min) (You et al., 2021b). However, a longer duration was associated with a reduced spontaneous respiratory frequency immediately after the exercise, which may be linked to positive psychophysiological outcomes (Narkiewicz et al., 2006; Nicolo et al., 2020). Regarding the inhalation/exhalation ratio, a longer exhalation than inhalation time was found to be related to increased CVA (Bae et al., 2021; Laborde et al., 2021c; Van Diest et al., 2014).

Two studies have focused on assessing the influence of a range of SPB respiratory frequencies on CVA (Lin et al., 2014; Song & Lehrer, 2003). Song and Lehrer (2003) examined respiratory rates of 3, 4, 6, 8, 10, 12, and 14 cpm. They found that the highest RSA amplitudes (operationalized as respiratory-induced trough heart rate) were achieved with 4 cpm, potentially reflecting the slow rate of acetylcholine metabolism. They concluded that respiratory rate affects the amplitude of RSA, with RSA amplitude increasing as respiration rate decreases (Brown et al., 1993; Cooke et al., 1998; Ritz et al., 2001; Stark et al., 2000). These increases are observed until 3 cpm, where RSA amplitude decreases again due to negative resonance (Song & Lehrer, 2003). Some limitations should be noted for this study, such as the small sample (n = 5), all female, and the inconsistent protocol used across participants (different number of sessions / wash-out times between respiratory frequencies). Further, the interpretation of the findings may prove challenging, as the amplitude of RSA does not necessarily reflect proportional changes in CVA (Kollai & Mizsei, 1990). In the second study, Lin et al. (2014) investigated four breathing patterns: 5.5 cpm and 6 cpm, with either a 5:5 or 4:6 inhalation/exhalation ratio. They found that the 5.5 cpm pattern with 5:5 inhalation/exhalation ratio triggered the highest increase in LF-HRV; however findings may have been influenced by an interaction between breathing frequency and inhalation/exhalation ratio, and only two different respiratory frequencies have been investigated (5.5 cpm and 6 cpm). Therefore, the current study keeps the inhalation/exhalation ratio constant and examines a larger range of SPB respiratory frequencies.

In summary, this study aimed to address the influence of several SPB respiratory frequencies on CVA. A within-subject design was used, as recommended to decrease inter-individual variability in HRV (Laborde et al., 2017b; Quintana & Heathers, 2014). We hypothesized that the control condition will trigger a lower CVA in comparison to all other SPB conditions. Given the mixed findings in previous research (Lin et al., 2014; Song & Lehrer, 2003), no specific predictions are made regarding the SPB respiratory frequency triggering the highest increase in CVA.

Methods

Participants

To determine the sample size for this study, we assumed a small-to-moderate effect size (f = 0.15) for the effects of SPB respiratory frequencies on CVA, based on previous research (Lin et al., 2014; Song & Lehrer, 2003). A G*Power (Faul et al., 2009) a priori power calculation with an effect size $f = 0.12$, Power (1-β) = 0.8, six measurements, and correlation among repeated measured = 0.50, provided an estimated sample size of 63. To account for potential dropouts and technical issues, a sample size of n = 78 was recruited. Following current recommendations regarding HRV studies (Laborde et al., 2017b), exclusion criteria were self-reported cardiovascular diseases, and other chronic diseases that might influence breathing or HRV patterns, such as asthma, diabetes, psychiatric, and neurological diseases. Due to technical issues, the data of three participants had to be excluded, and thus the final sample size comprised N = 75 participants (22 female; *M*age = 22,32; *SD* = 2,11; age range = 19–31); weekly sport participation (hours): 5.8 (*SD* = 2,9); BMI = 22.93 (*SD* = 2.33); waist-to-hips ratio = 0.85 (*SD* = 0.10). The protocol of the study was approved by the Ethics committee of the German Sport University Cologne (Number 136/2015).

Material and Measures

Cardiac Vagal Activity

An electrocardiography (ECG) device (Faros 180°, Bittium, Oulu, Finland) was used to measure HRV, with a sampling rate of 500 Hz. We used two disposable ECG pre-gelled electrodes (Ambu L-00-S/25, Ambu GmbH, Bad Nauheim, Germany). The negative electrode was placed in the right infraclavicular fossa (just below the right clavicle), while the positive electrode was placed on the left side of the chest, below the pectoral muscle in the left anterior axillary line. RMSSD and LF-HRVms^2 were extracted with Kubios (University of Eastern Finland, Kuopio, Finland). The full ECG recording was inspected visually, and artifacts were corrected manually (Laborde et al., 2017b). The focus was on vmHRV, which was indexed via RMSSD, and by LF-HRVms^2 (0.04 to 0.15 Hz). In order to provide an overview of the different HRV parameters, following recommendations in HRV research (Laborde et al., 2017b), we also extracted additional HRV parameters: heart rate, the standard deviation of the NN interval (SDNN) for the time-domain, and HFms^2 (0.15 to 0.40 Hz), and the LF/HF ratio were obtained for the frequency-domain (via Fast Fourier Transform).

Slow-Paced Breathing

SPB was performed with a video showing a little ball moving up and down at the rate of 5 cpm; 5.5 cpm; 6 cpm; 6.5 cpm; and 7 cpm. The inhalation / exhalation ratio was kept constant at 45% / 55%, a longer exhalation providing a stronger increase in cardiac vagal activity (Bae et al., 2021; Laborde et al., 2021c; Van Diest et al., 2014). The videos were recorded using the software EZ-Air Plus (Thought Technology Ltd., Montreal, QC, Canada), and were similar to the videos used in previous SPB experiments (e.g., Laborde et al., 2017a; You et al., 2021a). Participants were instructed to inhale continuously through the nose while the ball was moving upwards, and exhale continuously through the mouth while the ball was moving downwards. Breathing in through the nose not only helps to warm, clean, and moisten the air (Lorig, 2011), but it also triggers certain rhythmic patterns in our respiratory system. These patterns can stimulate synchronized electrical activity in the brain areas related to our sense of smell (piriform cortex) and areas connected to emotions and memory, such as the amygdala and hippocampus (Biskamp et al., 2017; Maric et al., 2020; Zelano et al., 2016). This suggests that nasal inhalation might contribute to an optimal level of activation in brain networks involved in processing stimuli and guiding our behavior. On the other hand, exhaling is usually done through the mouth, a process that offers less resistance to airflow than exhaling through the nose (Lorig, 2011). Further, exhaling with pursed lips allows for better control over the airflow, making it easier to adjust the exhalation duration (Fagevik Olsen et al., 2015; Noble & Hochman, 2019). This technique can also help to lower the respiratory rate. The exact instructions and detailed rationale for performing SPB this way can be found in previously published studies (e.g., Laborde et al., 2021a; You et al., 2021b).

Control Condition

The control condition, in which the participants were asked to breathe spontaneously, used an emotionally neutral TV documentary about world travel destinations, already used in previous research (e.g., Laborde et al., 2019b). The reasoning behind our selection of a TV documentary as the control condition stems from the consideration of the athletes' typical daily routines and habits. It is common for athletes to engage in activities such as watching documentaries or other videos on their TVs, smartphones, or laptops post-training. As such, this control condition not only reflects their real-life recreational activities but also ensures a relatable and familiar environment, minimizing the potential influence of novelty on the results.

Procedure

Participants were recruited via flyers on the campus of the local university and via posts on social network groups linked to the local university. Participants had to attend one testing session in the lab, where they performed the six breathing conditions in a randomized order. Each breathing condition lasted 5 min and was followed by a 5 min washout period. Randomization was performed using Research Randomizer (Urbaniak & Plous, 2013). Prior to the testing sessions, participants were instructed not to drink or eat anything but water for 2 h before the experiment, not to take part in any strenuous exercise or drink alcohol for the 24 h prior to testing (Laborde et al., 2017b). Upon arrival at the laboratory, participants were asked to fill out an informed consent form and a demographic questionnaire regarding variables potentially influencing HRV (Laborde et al., 2017b). At the beginning of the testing session, participants received a short video introduction on how to perform the technique correctly, which was checked by the experimenter. Additionally, the experimenter was checking visually whether the participants were following the timing of the breathing pacer when performing the different respiratory frequencies conditions, and if they were continuously inhaling and exhaling for the full duration of respectively the inhalation and exhalation periods. When small deviations occurred, the participant was instructed to follow the respiratory pattern more closely.

The ECG device was attached, and HRV was measured continuously throughout the experiment, in a sitting position, knees at 90°, hands on the thighs. All measures were conducted with eyes open. An overview of the protocol can be seen in Fig. 1.

Data Analysis

The ECG signal was imported into Kubios, and HRV variables were exported from the Kubios output. Data were checked for normality and outliers. Regarding outliers, 0.008% of the cases were found to be univariate outliers (>2 SD, z-scores higher than 2.58; none were found being >3 SD, with z-scores higher than 3.29) (Tabachnick & Fidell, 2012). Running the analysis without the outliers did not change the pattern of results, thus they were kept in the analysis. As the data were non-normally distributed, a log-transformation was applied, as it is usually recommended for HRV research (Laborde et al., 2017b).

We conducted a repeated-measures MANOVAs with Greenhouse-Geisser correction, with condition (5 cpm, 5.5 cpm, 6 cpm, 6.5 cpm, 7 cpm) as independent variable, and log RMSSD and LF-HRVms^2 as dependent variables. Age, sex, BMI, and waist-to-hips ratio were entered as covariates. Post-hoc *t*-tests comparisons were run with Bonferroni correction to further explore the significant effects, reporting the effect size with Cohen's d and the significance level. In total 30 t-tests were run (15 for log RMSSD and 15 for LF-HRVms^2), so we adjusted alpha to $0.05 / 30 = 0.0016$; and will set conservatively our significance level to 0.001.

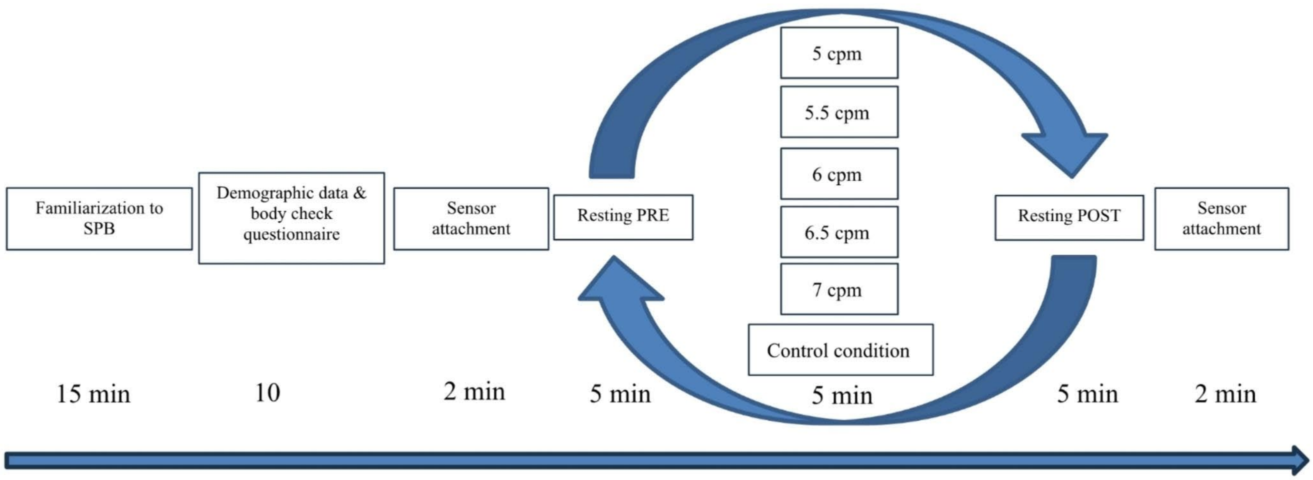


Fig. 1a Experimental protocol

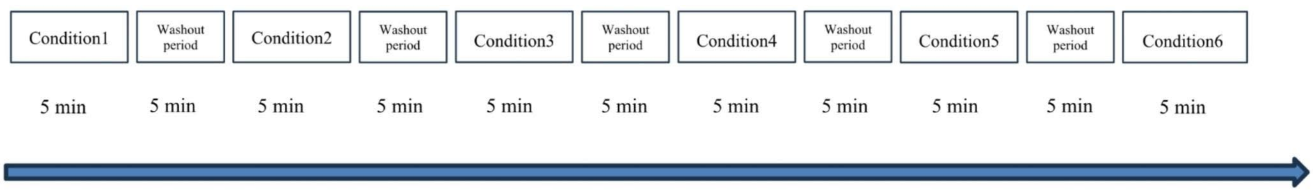


Fig. 1b Detailed overview of experimental conditions

Table 1 Descriptive statistics for all study variables

		5 cpm	5.5 cpm	6 cpm	6.5 cpm	7 cpm	Control
R-R interval	*M*	923,92	924,83	918,20	912,91	907,00	939,99
	SD	139,02	140,65	140,55	136,90	138,85	188,06
SDNN	*M*	150,67	148,19	141,37	135,80	126,85	98,23
	SD	43,53	44,47	45,35	44,21	42,90	40,70
Heart rate	*M*	68,16	68,07	68,46	68,66	68,99	65,62
	SD	10,12	10,30	10,45	10,08	10,31	10,45
RMSSD	*M*	93,79	97,77	97,62	95,63	91,72	62,84
	SD	40,50	42,02	43,81	42,74	42,55	35,96
Log RMSSD	*M*	1,93	1,94	1,94	1,93	1,91	1,73
	SD	0,22	0,23	0,23	0,23	0,22	0,24
LF_MS^2	*M*	19010,32	18797,61	17042,29	15520,09	13246,50	4045,44
	SD	10733,27	10515,76	10090,14	9576,70	9046,75	5177,40
Log_LF_MS^2	*M*	4,19	4,18	4,13	4,09	4,02	3,27
	SD	0,32	0,33	0,34	0,34	0,33	0,66
HF_MS^2	*M*	2241,98	2190,26	1852,22	1631,85	1206,86	1405,70
	SD	1945,21	1814,16	1684,75	1540,99	1206,39	1666,15
LF/HF	*M*	13,32	14,02	14,24	15,44	18,21	4,25
	SD	9,50	10,40	9,69	11,37	12,81	4,30

Note: SDNN = standard deviation of the R-R intervals, RMSSD = root mean square of the successive differences, LF = low frequency, HF = high frequency, cpm = cycles per minute

Results

Descriptive statistics are presented in Table 1 for all study variables, and mean comparisons regarding the post-hoc t-tests are shown in Tables 2 and 3.

Log RMSSD

For log RMSSD (see Fig. 2), we found a significant main effect of condition, with $F(2.404, 177.873) = 60.295$, $p < .001$, partial $\eta 2 = 0.45$. Running the analysis integrating the covariates, they were not found to have a significant

Table 2 Post-hoc t-tests for log RMSSD

Mean comparison	t	p	Cohen's d
Control vs. 5 cpm	9.785	< 0.001	-1.130
Control vs. 5.5 cpm	9.683	< 0.001	-1.118
Control vs. 6 cpm	9.746	< 0.001	-1.125
Control vs. 6.5 cpm	9.290	< 0.001	-1.073
Control vs. 7 cpm	8.555	< 0.001	-0.988
5 cpm vs. 5.5 cpm	1.595	0.115	-0.184
5 cpm vs. 6 cpm	1.286	0.203	-0.148
5 cpm vs. 6.5 cpm	0.356	0.723	-0.041
5 cpm vs. 7 cpm	1.061	0.292	0.123
5.5 cpm vs. 6 cpm	0.340	0.734	0.039
5.5 cpm vs. 6.5 cpm	1.159	0.250	0.134
5.5 cpm vs. 7 cpm	2.389	0.019	0.276
6 cpm vs. 6.5 cpm	0.939	0.351	0.108
6 cpm vs. 7 cpm	2.290	0.025	0.264
6.5 cpm vs. 7 cpm	1.474	0.145	0.170

Note: $df = 74$; cpm = cycles per minutes

Table 3 Post-hoc t-tests for log LF-HRVms^2

Mean Comparison	t	p	Cohen's d
Control vs. 5 cpm	12.637	< 0.001	-1.459
Control vs. 5.5 cpm	12.090	< 0.001	-1.396
Control vs. 6 cpm	11.332	< 0.001	-1.309
Control vs. 6.5 cpm	10.897	< 0.001	-1.258
Control vs. 7 cpm	10.006	< 0.001	-1.155
5 cpm vs. 5.5 cpm	0.519	0.605	0.060
5 cpm vs. 6 cpm	3.168	0.002	0.366
5 cpm vs. 6.5 cpm	5.210	< 0.001	0.602
5 cpm vs. 7 cpm	8.144	< 0.001	0.940
5.5 cpm vs. 6 cpm	3.639	< 0.001	0.420
5.5 cpm vs. 6.5 cpm	5.478	< 0.001	0.633
5.5 cpm vs. 7 cpm	8.359	< 0.001	0.965
6 cpm vs. 6.5 cpm	2.360	0.021	0.272
6 cpm vs. 7 cpm	5.830	< 0.001	0.673
6.5 cpm vs. 7 cpm	4.183	< 0.001	0.483

Note: $df = 74$; cpm = cycles per minutes

influence on the results: age ($p = .188$), sex ($p = .110$), BMI ($p = .321$), Waist-to-hips ratio ($p = .445$).

Post-hoc *t*-tests showed that the control condition differed significantly from all other conditions, 5 cpm ($d = 1.13$), 5.5 cpm ($d = 1.12$), 6 cpm ($d = 1.125$), 6.5 cpm ($d = 1.07$) = 1.07, 7 cpm ($d = 0.99$). No further mean comparisons between conditions were found to be significant.

Log Low-frequency

For log LF-HRVms^2 (see Fig. 3), we found a significant main effect of condition, with $F(1.298, 96.030) = 119.941$, $p < .001$, partial $\eta 2 = 0.62$. Running the analysis integrating the covariates were not found to have a significant influence on the results: age ($p = .143$), sex ($p = .223$), BMI ($p = .447$), Waist-to-hips ratio ($p = .910$).

Post-hoc t-tests showed that the control condition was significantly lower than all other conditions, 5 cpm ($d = 1.46$), 5.5 cpm ($d = 1.40$), 6 cpm ($d = 1.31$), 6.5 cpm ($d = 1.26$) = 1.073, 7 cpm ($d = 1.16$). The 5 cpm condition was significantly higher than the 6.5 cpm condition ($d = 0.602$) and the 7 cpm condition ($d = 0.94$). The 5.5 cpm condition was higher than the 6 cpm condition ($d = 0.42$), 6.5 cpm condition ($d = 0.63$), 7 cpm condition ($d = 0.97$). The 6 cpm condition was higher than the 7 cpm condition ($d = 0.67$). The 6.5 cpm condition was higher than the 7 cpm condition ($d = 0.48$). No further mean comparisons between conditions were found to be significant.

Discussion

The aim of this study was to investigate the effects of several SPB respiratory frequencies on CVA, as indexed by RMSSD and LF-HRVms^2. In line with our hypothesis, all SPB respiratory frequencies triggered a higher CVA in comparison to the control condition (spontaneous breathing). Further post-hoc analyses revealed no differences between SPB respiratory-frequencies for RMSSD, while differences appeared for LF-HRVms^2. Specifically, LF-HRVms^2 was found to be higher in four conditions (5 cpm, 5.5 cpm, 6 cpm, and 6.5 cpm) when compared to 7 cpm; higher in the 5 cpm condition compared to the 6.5 cpm condition; and higher in the 5.5 cpm condition compared to the 6 cpm and 6.5 cpm conditions. These findings are discussed below.

The finding that SPB increases vmHRV is in line with the findings of a recent meta-analysis (Laborde et al., 2022a). As evidenced by a pharmacological blockade study focusing on LF-HRVms^2 (Kromenacker et al., 2018), the increase observed in LF-HRVms^2 is suggested to reflect the activation of the vagus nerve. Several mechanisms may be at stake, spanning the improvement of baroreflex sensitivity, the increase of amplitude of RSA, the stimulation of pulmonary stretch receptors, and the stimulation of brain networks involved in emotion regulation (Gerritsen & Band, 2018; Kromenacker et al., 2018; Lehrer & Gevirtz, 2014; Mather & Thayer, 2018; Noble & Hochman, 2019; Sevoz-Couche & Laborde, 2022).

Aside from spontaneous breathing, no differences were found among SPB respiratory frequencies with RMSSD. However, differences were observed with LF-HRVms^2, this may illustrate that the absolute power of LF-HRV may be more sensitive to changes in SPB respiratory frequencies than RMSSD. The recent meta-analysis of Laborde et al. (2022a) showed that SPB triggered moderate increases in RMSSD (Hedge's g = 0.53), while it triggered large increases in LF-HRV (Hedge's g = 1.49). This difference in magnitude of effects may therefore explain the difference in sensitivity of

Fig. 2 RMSSD across breathing conditions. Note: * = 0.01 < $p \leq .05$, ** = 0.001 ≤ $p \leq .01$, *** = $P < .001$, cpm = cycles per minute

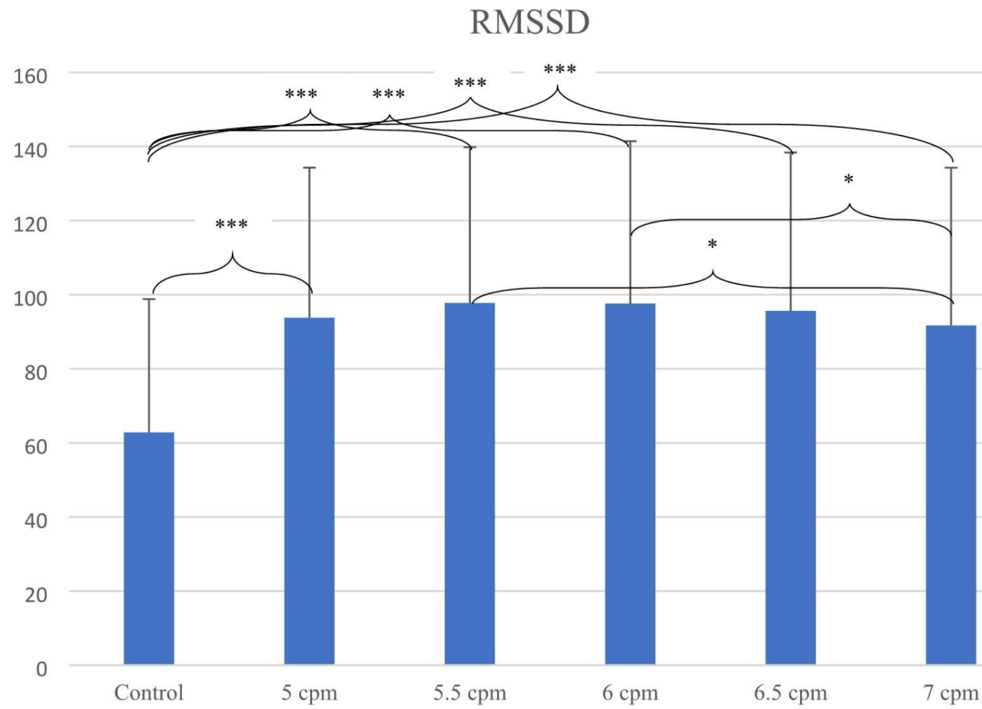
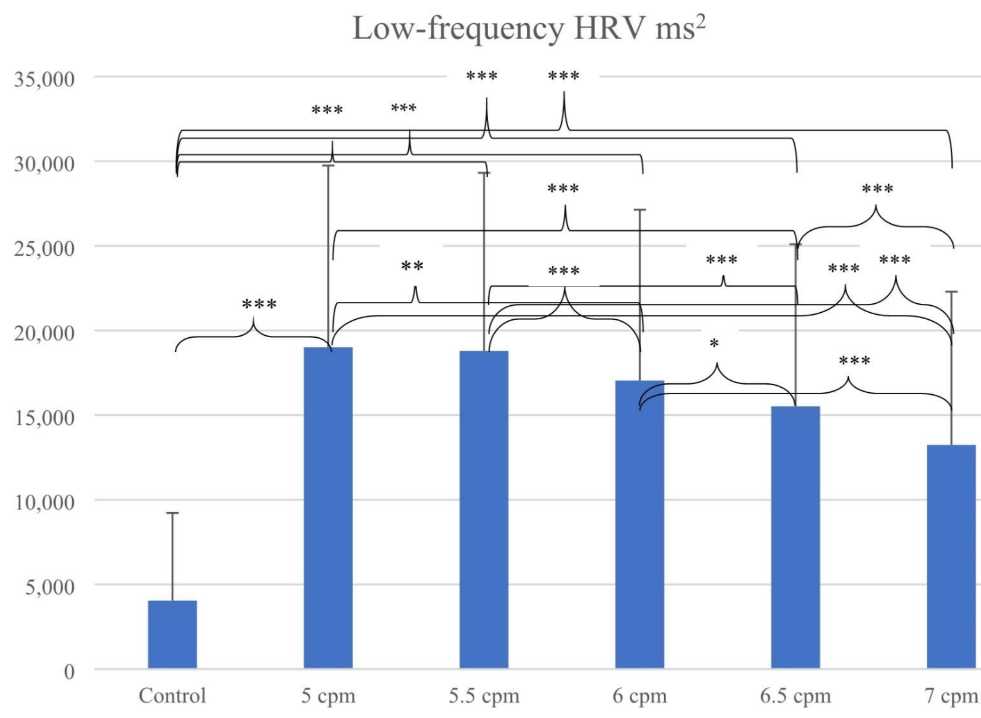


Fig. 3 LF-HRV ms^2 across breathing conditions. Note: * = 0.01 < $p \leq .05$, ** = 0.001 ≤ $p \leq .01$, *** = $P < .001$, cpm = cycles per minute



the two variables. This is potentially due to LF-HRV reflecting maximized baroreflex sensitivity and RSA amplitude as respiratory frequency approaches the resonance frequency (Lehrer et al., 2000; Shaffer & Meehan, 2020). In addition, the resonance frequency is typically determined through LF-HRV power and RSA amplitude (heart rate max-min) for each breath cycle (Lehrer et al., 2000; Shaffer & Meehan, 2020), so further research should also investigate changes in RSA amplitude for each of the respiratory frequencies, in addition to RMSSD and LF-HRV. Furthermore, the difference in the sensitivity of LF-HRVms^2 and RMSSD to SPB may be due to their calculation methods. While computing RMSSD involves the comparison of successive beat-to-beat changes, the computation of LF-HRVms^2 relies on power spectrum analysis (Berntson et al., 1997; Malik, 1996). As vagal influences may occur within one second (Borst & Karemaker, 1983), variations in adjacent RR intervals are thought to exclusively represent vagal influences, being less influenced by respiratory frequency (Goedhart et al., 2007; Hill et al., 2009; Penttila et al., 2001; Quintana et al., 2016) – but see also (Berntson et al., 2005). Given the pharmacological blockade study of Kromenacker et al. (2018) only reported LF-HRVms^2 and not RMSSD, no direct comparison between both indicators was realized under SPB conditions. Previous pharmacological blockade studies not involving SPB showed that RMSSD is a reliable marker for cardiac parasympathetic activity (Goldberger et al., 2006; Ng et al., 2009) and that RMSSD correlates with HRV in the frequency-domain in non-SPB condition (i.e., HF-HRV) (Ng et al., 2009). To better understand the relationship between SPB respiratory frequencies and the diverse vmHRV indicators, future pharmacological studies are recommended to go beyond previous research (Hayano et al., 1994; Kromenacker et al., 2018). This would involve double blockade (i.e., sympathetic and parasympathetic) with a systematic manipulation of respiratory frequency, as well as potentially of other respiratory parameters, with the report of both time and frequency-domain HRV indices, as pointed out by HRV recommendations (Laborde et al., 2017b).

The finding that lower SPB respiratory frequencies triggered higher increases in LF-HRVms^2 may be related to previous findings regarding the resonance frequency (Vaschillo et al., 2006). Indeed, our sample mostly comprised male participants (71%), and Vaschillo et al. (2006) found a lower resonant frequency in men than women, given men are suggested to have a larger vascular tree than women. In addition, Vaschillo et al. (2006) found a negative relationship between resonant frequency and height, but no relationship between resonant frequency and age, weight, or presence of asthma. It was not the aim of this study to determine the exact resonance frequency of each participant, and as such the resonance frequency determination protocols (Fisher & Lehrer, 2021; Lehrer et al., 2000, 2013; Shaffer & Meehan, 2020) have not been followed, but may be considered in future research.

Our study had several strengths, for example being the first study investigating several SPB respiratory frequencies with CVA, but also some limitations. First, we adopted a group average approach and did not consider an approach determining the individual's resonance frequency, as mentioned above (Shaffer & Meehan, 2020; Vaschillo et al., 2002). In line with this consideration, our study investigated SPB as low as 5 cpm. However, given resonance frequency may go below 5 cpm (Lehrer et al., 2000; Shaffer & Meehan, 2020), future research should consider first determining resonance frequency, and then investigate its influence on vmHRV compared to standard SPB frequencies. Second, this study was focusing on vmHRV, and did not measure any other physiological measure, such the end-tidal volume, oxygen saturation, carbon dioxide, or baroreflex sensitivity (e.g., Cooke et al., 1998). These measures should be considered in future research, to achieve a more comprehensive overview of the physiological effects of different SPB respiratory frequencies. Third, we only focused on manipulating respiratory frequency, while other respiratory parameters may also influence vagal outflow, such as the inhalation/exhalation ratio, with a longer exhalation triggering higher increases in CVA (Bae et al., 2021; Laborde et al., 2021c; Van Diest et al., 2014), or postinspiratory and postexpiratory periods, with CVA suggested to be enhanced during the postinspiratory period and inhibited during the postexpiratory period (Spyer & Gilbey, 1988). Fourth, future research should endeavor to investigate whether findings generalize to different populations (healthy vs. clinical), considering different demographical aspects (age, sex, weight, height etc.). Sixth, SPB training with and without biofeedback is typically conducted over the long term to investigate its influence on long-term outcomes. However, the current study focused on acute effects. Further research should therefore consider adopting a long-term perspective to better understand how vmHRV variables are influenced by changes in SPB frequencies over an extended period. Finally, future research must consider whether changing SPB respiratory frequency may also translate into influencing specific outcomes beyond impacting CVA. For example outcomes that are traditionally associated with the positive impact of practicing slow-paced breathing (Lehrer et al., 2020) such as stress management and emotional regulation.

Conclusion

The results of the present study show that any SPB respiratory frequency between 5 cpm and 7 cpm would trigger an increase in CVA, in comparison to a spontaneous respiratory frequency, in line with the findings of a recent meta-analysis (Laborde et al., 2022a). This result is of utmost importance for athletes, when there is no time to define the individual resonance frequency (Fisher & Lehrer, 2021; Lehrer et al., 2000, 2013; Shaffer & Meehan, 2020). This could mean that brief training for athletes on slowing their breathing between 5 and 7 cpm can enhance CVA and potentially positive self-regulatory outcomes for performance.

In addition to SPB (Laborde et al., 2022a), several interventions exist to enhance CVA (Laborde et al., 2018), such as with the diving reflex (Ackermann et al., 2022), or with non-invasive brain stimulation (Schmaußer et al., 2022). Future research may endeavor to investigate further possible interactions between CVA enhancing techniques, given the association between CVA and several psychophysiological states relevant for sport performance, such as stress management and emotion regulation (Mosley & Laborde, 2022). The application of brief SPB interventions, which can be delivered through accessible devices like smartphones (Mosley et al., 2023), appears as a promising avenue to further explore for both researchers, athletes and practitioners in sport and exercise psychology.

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Data Availability Data supporting the study findings can be provided by the corresponding author, Min You, upon reasonable request.

Declarations

Competing interest Authors declare that there are no financial conflicts of interest related to the research reported in this manuscript. The authors have no conflicts of interest, and the research did not receive any external funding.

Ethics Approval The study protocol was approved by the local university's Ethics Committee (Approval Number 136/2015) and conducted in accordance with the Declaration of Helsinki. Informed consent was obtained from all participants involved in the study.

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